

CHANDRA SHEKHAR SINGH

Ph.D., Mechanical Engineering

1615/2A, Tubewell Road 3, Jhansi, Uttar Pradesh – 284003, India

+91-6005543423 | cssbisht@gmail.com | cssbisht_12phd18@nitsri.ac.in



EDUCATION

- ❖ **Ph.D, Mechanical Engineering Department, National Institute of Technology Srinagar, Jammu & Kashmir, India.** 2024
CGPA: 8.307. Dissertation: *Synthesis and Microstructural, Physical, Mechanical, and Tribological Characterization of Cu-Ni Alloy Based Metal Matrix Composites for Railway Applications.*
Supervisor: Prof. Rakesh Sehgal (HAG, Former Director, NIT Srinagar) & Prof. M. F. Wani (HAG)
- ❖ **M.Tech, Production Engineering Department, Guru Nanak Dev Engineering College, Ludhiana, Punjab, India.** 2011
Grade: 66.09%. Thesis: *Strength Investigation of Cruciform Joint using Finite Element Modelling.*
Supervisor: Dr. Gurinder Singh Brar (Currently Associate Professor, NIT Uttarakhand)
- ❖ **B.Tech, Mechanical Engineering Department, Institute of Engineering & Technology, Bundelkhand University Jhansi, Uttar Pradesh, India, Grade: 67.26%.** 2008
- ❖ **Intermediate (12th), Government Intermediate College, Jhansi, Uttar Pradesh, India** 2004
Uttar Pradesh Board, Lucknow, India, Grade: 56.8%.
- ❖ **High School (10th), St. Mary's High School, Jhansi, Uttar Pradesh, India** 2002
Uttar Pradesh Board, Lucknow, India, Grade: 62.33%.

JOURNAL PUBLICATIONS

- ❖ **Chandra Shekhar**, M. F. Wani, Rakesh Sehgal, Umida Ziyamukhamedova, and Nodirjon Tursunov. "A novel ceramic reinforced metal matrix composite (Cu-Ni/TiC-CaF₂): Fabrication, Microstructure, Mechanical and Tribological Characterization", *Metallurgical And Materials Transactions B*. Publisher: Springer, Indexed in: SCIE, Scopus, EI Compendex. **Impact Factor: 2.6**, Quartile/Category: Q1 / Materials Science, Q2 / Materials Chemistry. <https://doi.org/10.1007/s11663-024-03418-2>.
- ❖ M Kocak, S Kumar, M Mrlik, B Kumar, **Chandra Shekhar**, R Sehgal, M F Wani, M Kubik, Michal Sedlacik. "Tribological characteristics of magnetorheological fluids based on carbonyl iron particles coated with various types of organosilanes", *Journal of Materials Research and Technology* 36 (2025) 5962–5977. Publisher: Elsevier, Indexed in: SCIE, Scopus. **Impact Factor: 6.2**, Quartile/Category: Q1 / Metallurgy & Metallurgical Engineering. <https://doi.org/10.1016/j.jmrt.2025.04.238>.
- ❖ S Kumar, V. Singh, B. Kumar, H. S. Gupta, **Chandra Shekhar**, R Sehgal, M F Wani, M. D. Sharma, M. Sedlačk and Khosla. "Investigating the Lubrication Properties of PAO-4 Oil with Hybrid GO/Fe₂O₃ Nanocomposite Additives", *Journal of Tribology*, TRIB-24-1506, 1-31, (2025). Publisher: ASME, Indexed in: SCIE, Scopus. **Impact Factor: 2.2**, Quartile/Category: Q2 / Mechanical Engineering. doi: <https://doi.org/10.1115/1.4068781>.
- ❖ **Chandra Shekhar**, M. F. Wani, Rakesh Sehgal, Sheikh Shahid Saleem, Umida Ziyamukhamedova, and Nodirjon Tursunov. "Recent Progress in Particulate Reinforced Copper-Based Composites: Fabrication, Microstructure, Mechanical, and Tribological Properties—A Review", *Advanced Engineering Materials*, (2024) 2401748. Publisher: Wiley-VCH Verlag, Indexed in: SCIE, Scopus. **Impact Factor: 3.4**, Quartile/Category: Q1 / Condensed Matter Physics, Q2 / Materials Science. <https://doi.org/10.1002/adem.202401748>.
- ❖ **Chandra Shekhar**, M. F. Wani, and Rakesh Sehgal. "Effect of CaF₂ nanoparticles on the microstructure evolution, mechanical properties and tribological behaviour of Cu-Ni alloy/cBN self-lubricating composites", *Tribology International* 188 (2023): 108884. Publisher: Elsevier, Indexed in: SCIE, Scopus. **Impact Factor: 6.1**, Quartile/Category: Q1 / Mechanical Engineering, Q1 / Mechanics of Materials, Q1 / Surfaces and Interfaces. <https://doi.org/10.1016/j.triboint.2023.108884>.
- ❖ **Chandra Shekhar**, M. F. Wani, and Rakesh Sehgal. "Fabrication and self-lubricating tribological characterisation of Cu-Ni/TiC/CaF₂ composite for railway switch slide baseplate", *Wear* 526 (2023): 204890. Publisher: Elsevier, Indexed in: SCIE, Scopus. **Impact Factor: 5.3**, Quartile/Category: Q1 / Condensed Matter Physics, Q1 / Materials Chemistry, Q1 / Mechanics of Materials, Q1 / Surfaces and Interfaces. <https://doi.org/10.1016/j.wear.2023.204890>.
- ❖ Gurinder Singh Brar and **Chandra Shekhar Singh**, "FEA of residual stress in cruciform welded joint of hollow sectional tubes", *Journal of Constructional Steel Research*, 102 (2014): 44-58. Publisher: Elsevier, Indexed in: SCIE, Scopus. **Impact Factor: 4.0**, Quartile/Category: Q1 / Mechanics of Materials, Q1 / Metals and Alloys, Q1 / Building and Construction. <https://doi.org/10.1016/j.jcsr.2014.06.011>.
- ❖ Pranav Dev Srivyas, M. F. Wani, Rakesh Sehgal, **Chandra Shekhar Singh Bisht**, M. S. Charoo, Ankush Raina, and Mir Irfan Ul Haq. "Synergetic effect of surface texturing and graphene nanoplatelets on the tribological properties of hybrid self-lubricating composite", *Tribology International* 168 (2022): 107434. Publisher: Elsevier, Indexed in: SCIE, Scopus. **Impact Factor: 6.1**,

Quartile/Category: Q1 / Mechanical Engineering, Q1 / Mechanics of Materials, Q1 / Surfaces and Interfaces.
<https://doi.org/10.1016/j.triboint.2022.107434>.

- ❖ Pranav Dev Srivyas, M. S. Charoo, M. F. Wani, Rakesh Sehgal, Ankush Raina, Mir Irfan Ul Haq, **Chandra Shekhar**, Tanmoy Medhi, and Soundhar Arumugam. "Impact of surface texturing on the tribological behaviour of aluminium-silicon (Al-Si/Al2O3) advanced composite under dry and lubricating conditions", *Surface Topography: Metrology and Properties* 10, no. 3 (2022): 035043. Publisher: IOP Publishing, Indexed in: SCIE, Scopus. **Impact Factor: 2.0**, Quartile/Category: Q2 / Materials Chemistry, Q2 / Surfaces, Coatings and Films. [DOI 10.1088/2051-672X/ac929a](https://doi.org/10.1088/2051-672X/ac929a).
- ❖ Ziyamukhamedova, Umida, Shah Wasi, Sanjay Kumar, Rakesh Sehgal, M. F. Wani, **Chandra Shekhar Singh**, Nodirjon Tursunov, and Himanshu Shekhar Gupta. "Investigating Friction and Antiwear Characteristics of Organic and Synthetic Oils Using h-BN Nanoparticle Additives: A Tribological Study", *Lubricants* 12, no. 1 (2024): 27. Publisher: MDPI, Indexed in: SCIE, Scopus. **Impact Factor: 3.1**, Quartile/Category: Q2 / Mechanical Engineering, Q2 / Surfaces, Coatings and Films. <https://doi.org/10.3390/lubricants12010027>.
- ❖ Jagtar Singh, M. F. Wani, Summera Bandy, **Chandra Shekhar**, and Gurtej Singh. "Nano scratch and nanoindentation: an approach to understand the tribological behaviour of max phase material Ti2AlC", *IOP Conf. Series: Materials Science and Engineering*, vol. 561, no. 1, p. 012111. IOP Publishing, 2019. [DOI 10.1088/1757-899X/561/1/012111](https://doi.org/10.1088/1757-899X/561/1/012111).

RESEARCH INTERESTS

Metal Matrix Composites, Tribology & Wear Analysis, Powder Metallurgy & Composite Fabrication, Laser Texturing, Coating & Surface Modification, Self-lubricating composites, Materials testing and characterization.

REVIEWER OF SCI/PEER REVIEWED JOURNALS

- **Engineering Structures- Elsevier** (SCIE, SSCI, Scopus, Web of Science Indexed, **Impact Factor: 5.6**).
- **Materials Research Express- IOP Publishing** (SCI, SCOPUS Indexed, **Impact Factor: 1.4**).
- **Nano- World Scientific Publishing** (SCI, SCOPUS Indexed, **Impact Factor: 1.5**).
- **Journal of Tribology- ASME** (SCI, SCOPUS Indexed, **Impact Factor: 2.2**).
- **Surface Topography: Metrology and Properties- IOP Publishing** (SCI, SCOPUS Indexed, **Impact Factor: 2.56**).
- **Engineering Research Express- IOP Publishing** (SCI, SCOPUS Indexed, **Impact Factor: 1.2**).
- **Scientific Reports- Springer Nature** (SCI, SCOPUS Indexed, **Impact Factor: 3.8**).

PROFESSIONAL EXPERIENCE

- **Tribology Laboratory, Mechanical Engineering Department, National Institute of Technology, Srinagar, J&K, India**
Innovative Research Fellow, Lab Handling & Academic Assistantship November 2018 to August 2024
 - ◆ Fabricated and characterized Cu-Ni alloy-based MMCs.
 - ◆ Conducted Physical, Mechanical and Tribological testing using specialized instruments.
 - ◆ Analysed ceramic-reinforced composites microstructure properties, utilized advanced characterization tools, including Leica Optical Microscope, XRD, FESEM with EDS, and Raman Spectrometer.
 - ◆ Delivered tutorials and laboratory sessions.
- **Ideal Institute of Management & Technology (Ideal Group of Institutions)** Ghaziabad, India
Assistant Professor, Department of Mechanical Engineering July 2013 to June 2017
 - ◆ Courses Taught at Graduate Level (B.Tech): Strength of Material, Machine Design, Fluid Mechanics, Computer Aided Manufacturing, Measurement Metrology and Control.
 - ◆ Examination Responsibilities: Worked as an Assistant Centre Superintendent in Diploma Examination of Board of Technical Education, Uttar Pradesh, and Working as an Exam Coordinator in Internal Examinations.
 - ◆ Timetable Coordinator: Scheduling all subjects taught in the B.Tech programme (Mechanical Engineering).
- **Anand Engineering College (Sharda Group of Institutions)** Agra, India
Assistant Professor, Department of Mechanical Engineering August 2012 to June 2013
 - ◆ Courses Taught at Graduate Level (B.Tech): Fluid Mechanics, Engineering Drawing and Workshop Practice.
 - ◆ Courses Taught at Post-Graduate Level (M.Tech): Metal Casting.
- **Institute of Engineering & Technology, Bundelkhand University Jhansi** Jhansi, India
Guest Lecturer, Department of Mechanical Engineering August 2011 to June 2012
 - ◆ Courses Taught at Graduate Level (B.Tech): Machine Design, Strength of Material, Computer Aided Manufacturing and Machine Drawing.
- **Institute of Engineering & Technology, Bundelkhand University Jhansi** Jhansi, India
Guest Lecturer, Department of Mechanical Engineering August 2008 to June 2009
 - ◆ Courses Taught at Graduate Level (B.Tech): Machine Design, Computer Aided Manufacturing and Applied Thermodynamics.

SOFTWARE & DATA ANALYSIS TOOLS AND RESEARCH EXPERTISE

- Solidworks, OriginPro 2023b and MS-Office.
- Cu- Metal Matrix Composites (MMCs) Fabrication & Characterization
- Understanding of microstructural development in MMCs
- Experimental Design and Reporting: Experimental Design, Result Interpretation, and Scientific Writing.

TECHNICAL SKILLS AND EQUIPMENT PROFICIENCY

❖ Proficient in operating various testing equipment:

<u>Fabrication & Prepration Equipments:</u> ◆ Vacuum Tube Furnace ◆ Planetary Ball Milling ◆ Molding Machine ◆ Automatic Polishing Machine ◆ Diamod Cutter Machine	<u>Characterization Tools:</u> ◆ Leica Optical Microscope ◆ 3D Profilometer ◆ Familiarity with XRD Machine ◆ Familiarity with FESEM with EDS ◆ Familiarity with Raman Spectrometer
<u>Mechanical Testing Instruments:</u> ◆ Vicker's Microhardness	<u>Surface Modification Machinery:</u> ◆ Laser Texturing Machine
<u>Tribological Testing Equipment:</u> ◆ Universal Tribometer (Reciprocating, Rotary, High Temperature, Tribocorrosion) ◆ Nano Tribometer	◆ Engine Tribometer (Equipped with a piston ring and liner reciprocating setup) ◆ Fourball Tribotesting instrument

PAPERS PRESENTED IN NATIONAL / INTERNATIONAL CONFERENCES

Conference	Authors, Title & Year
◆ International Conference on Tribology under the theme of Sustainable Development through Tribology (TRIBOINDIA-2023), Department of Mechanical Engineering, National Institute of Technology Srinagar, Hazratbal, Srinagar, India	Chandra Shekhar, "A Synergic Effect of TiC and CaF ₂ Reinforcement on Tribological Behaviour of Cu-Ni Matrix based Composite". (October 5-7 th 2023).
◆ International Conference on Tribology under the theme of Sustainable Development through Tribology (TRIBOINDIA-2023), Department of Mechanical Engineering, National Institute of Technology Srinagar, Hazratbal, Srinagar, India	Chandra Shekhar, "Nano-Indentation and Tribological Behaviour of Cu-Ni Alloy Synthesized by Powder Metallurgy". (October 5-7 th 2023).
◆ National Conference on Innovation in Futuristic Materials & Manufacturing Techniques. IFMMT-2014, School of Engineering & Technology, Faculty of Engineering, Manipal University Jaipur, India.	Chandra Shekhar, "Distribution of Residual Stresses in Different Parts Produced by Different Manufacturing Processes and its Measurement Techniques: A Comprehensive Review". (December 26 -27, 2014).
◆ National Conference on Recent Advances in Mechanical Engineering. NCRAME-2013, GBPEC Pauri Garhwal, Uttarakhand, India.	Chandra Shekhar, "Effect of the Welding Techniques on Residual Stresses Developed in Cruciform Weld Joint". (08 - 09 July 2013, Pages 155-160).

SHORT TERMS COURSES/ SYMPOSIUM /WORKSHOPS/ INDUSTRIAL TRAININGS

- ❖ Participated in "Pre-Conference Course" of TribolIndia-2023, an International Conference on Tribology, under the theme on "Nano Tribology & Thin Film Characterization" held on 4th October 2023, organized by Department of Mechanical Engineering, "National Institute of Technology Srinagar under the aegis of Tribology Society of India (TSI)", (Participation Certificate).
- ❖ Participated in one-week short-term course titled "Advanced Materials for Structural Applications", conducted as part of a quality improvement program by the Department of Metallurgy Engineering and Materials Science at the "Indian Institute of Technology, Indore, India", from 29th September to 4th October 2020, (Participation Certificate).
- ❖ Attended a one-week online short-term course on "Future Scope in Engineering Materials and Tribology", sponsored by TEQIP-III and organized by the Department of Mechanical Engineering at "Dr B R Ambedkar National Institute of Technology, Jalandhar, India", held from 15th to 19th September 2020, (Participation Certificate).
- ❖ Participated in one-week online workshop on "Nanotechnology and Nanomaterials for Interdisciplinary Applications (NNIA-2020)" organized by the Department of Mechanical Engineering at "Maulana Azad National Institute of Technology, Bhopal, India", from July 27th to 31st, 2020, (Participation Certificate).
- ❖ Attended a short-term online course on "Laser Micro Manufacturing, Surface & Material Processing", organized by "Indian Institute of Technology, Indore, India", from July 15th to 17th, 2020, (Participation Certificate).
- ❖ Participated in one day orientation workshop on Unnat Bharat Abhiyan organized by the National Institute of Technology, Srinagar, in 17th February 2020. (Participation Certificate)

- ❖ **Contributed** to organizing a one-week TEQIP III sponsored Faculty Development Program on "**Outcome-based Education**," organized by the Department of Mechanical Engineering, **National Institute of Technology, Srinagar**, from 18th to 22th May 2019. (Appreciation Certificate for Contribution as a member of the Organizing team)
- ❖ **Contributed** to organizing the 2nd International Indo-Tunisian Symposium-2019 on "**Tribology for Sustainability**", organized by the Department of Mechanical Engineering, **National Institute of Technology, Srinagar**, from 17th to 21th June 2019, (Appreciation Certificate for Contribution as a member of the Organizing team).
- ❖ **Participated** in One Day Workshop on "**Application of Lean Six Sigma in Indian Industry**", organized by Department of Industrial and Production Engineering, **Dr B R Ambedkar National Institute of Technology, Jalandhar** on 28th April 2012, (Participation Certificate).
- ❖ **Participated** in One Month Foundation Course on "**Solidworks-2009 - 3D Modeling**", at **CAD-DESK, Design software Training & Technical Institute, Jhansi**, from 7th November to 6th December 2011, (Participation Certificate).
- ❖ **Participated** in One Day Workshop on "**Patent Awareness**", Organized by **Guru Nanak Dev Engineering College, Ludhiana** in association with Patent Information Centre, on 18th December 2009, (Participation Certificate).
- ❖ **Attended** Two Weeks Training Course on "**CNC Horizontal Machining Centre (HMC-500) programming, operation & practice (Sinumeric 8400 system)**", at **Advanced Training Institute, Udyog Nagar, Kanpur**, from 05th to 16th January 2009.
- ❖ **Participated** in One Month Industrial Training on "**Machine Shop, Air Brake Lab, Mill Wright Shop & Welding Shop**", at **Railway Wagon Repair Shop, Jhansi**, from 15th June to 14th July 2007, (Participation Certificate).
- ❖ **Participated** in One month Industrial Training on "**Computer Numerical Control (CNC) Machine**", at **Bharat Heavy Electricals Limited, Jhansi**, from 25th July to 23rd August 2006, (Participation Certificate).

AWARDS & SCHOLARSHIPS

- ❖ Awarded TEQIP-III Scholarship and Institute Scholarship during Ph.D. at National Institute of Technology, Srinagar.
- ❖ Received Railway Employee Cooperative Credit Society Scholarship during B.Tech.

PERSONAL DETAILS

Father's Name: Mohan Singh Bisht
Category: General
Marital Status: Married

Date of Birth: August 24, 1986
Gender: Male
Language Known: Hindi (S/R/W), English (S/R/W)

REFERENCES

- **Prof. Rakesh Sehgal**
 HAG Professor, Mechanical Engineering Department, National Institute of Technology, Hamirpur, Himachal Pradesh, India – 177005, (Former Director, National Institute of Technology, Srinagar). ✉ rakeshsehgal.nitham@gmail.com | ☎ +91-9418058442 |
- **Prof. M. F. Wani**
 HAG Professor, Mechanical Engineering Department, National Institute of Technology, Srinagar, Kashmir, J&K, India – 190006. ✉ mfwani@nitsri.ac.in | ☎ +91-8803824243 |

I certify that the foregoing information is correct and complete to the best of my knowledge and belief and nothing has been concealed/distorted.

Dated:


 CHANDRA SHEKHAR